

Sectoral macroeconomic damage functions of weather extremes in Europe

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1 Introduction

Quantifying the economic damages stemming from climate change is crucial for designing effective climate policy and determining optimal investments in mitigation and adaptation. While climate encompasses a complex collection of atmospheric and environmental conditions that cannot be easily reduced to a few metrics, the climate econometrics literature has historically relied on variation in mean temperature as a proxy for climate change (*e.g.*, Dell et al. (2012); Burke et al. (2015)). This simplification has been pragmatic: temperature variation correlates well with the many dimensions of climate change and offers easy interpretation as well as the conciseness required by empirical analyses with limited statistical power.

More recently, researchers have argued that global variations in mean temperature, rather than country-level temperature changes, better capture the atmospheric alterations associated with climate change because they more effectively proxy for extreme weather events such as droughts, intense precipitation, or cyclones (Bilal and Känzig, 2024). However, this reduction of climate to simple proxies, while methodologically convenient, provides limited insight into the mechanisms through which climate, and climate change in particular, affects economic growth at the macroeconomic scale.

Understanding these channels may not be necessary for analyzing past impacts but is crucial for accurately modeling future ones. As economies evolve, they fundamentally change how they respond to climate conditions. The classic example is structural transformation: agriculture-based economies are typically more vulnerable to rainfall and temperature patterns than industrialized and service-oriented economies. Yet this conventional wisdom requires empirical validation, especially as climate change intensifies extreme weather events that may affect sectors in unexpected ways.

This work advances the literature in three key directions. First, following Akyapi et al. (2024), we employ a data-driven process to identify which climate variables best capture the variation in climate that matters for economic growth. However, we extend their approach by examining sector-specific effects, recognizing that different economic sectors may have distinct vulnerabilities to climate variables. Second, we leverage subnational data (Wenz et al., 2023), allowing us to capture within-country heterogeneity that might be masked in country-level analyses. Third, given

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that climate change is expected to increase the frequency and severity of extreme weather events, we focus expressly on such variables.

The regional disaggregation allows us to capture within-country heterogeneity that might be masked in country-level analyses. In addition, the larger sample size allows us to restrict the focus to a limited number of countries. Global analyses of the effects of weather shocks on economic growth have shown that rich, high-latitude countries are less vulnerable to climate change (Burke et al., 2015; Bilal and Känzig, 2024). However, global analyses typically force a quadratic relationship between temperature and economic growth that might poorly represent the dynamics in the tails of the distribution. There’s evidence indeed that positive temperature shocks have convex effects on growth in European regions with mild climate (Groom et al., 2023).

In the sections that follow, we describe our data sources and methodology, present our empirical results, and discuss their implications for climate policy and economic modeling.

2 Data

Weather data. We compile a dataset of weather extremes on a 0.25 degree grid from 1960 to 2020 using ERA5-Land reanalysis data. The full set of variables is described in Appendix A.1. Weather extremes are averaged at sub-national levels weighting by population density in 2015, from Center For International Earth Science Information Network-CIESIN-Columbia University (2018).

Economic data. Sub-national sectoral economic data comes from Wenz et al. (2023). Specifically, we use are interested in the total growth rate of gross regional product, and the growth rate of gross regional product of agriculture, manufacturing, and services. For a detailes description of the data, see Wenz et al. (2023).

3 Methodology

The relationship between weather variables and economic growth can be modeled as:

$$\Delta \log(y_{it}) = \beta_1 W_{it} + \lambda_i + \lambda_t + t \cdot \lambda_i + \epsilon_{it} \quad (1)$$

where W_{it} is a vector of weather variables, λ_i and λ_t are region and year fixed effects, and ϵ_{it} is the error term.

We estimate the effect of weather variables on the growth rate of gross regional product using a post-LASSO regression. We follow Akyapi et al. (2024) in using a LASSO regression to select the most relevant weather variables for predicting economic growth. We then estimate the effect of these variables on the growth rate of gross regional product using a post-LASSO regression.

First, we residualize all independent and dependent variables on region fixed effects, year fixed effects, and linear region-time trends to ensure fixed effects and trends are correctly included in the post-LASSO regression. We then standardize the residuals of the dependent variable and the independent variables. This is done to ensure that all variables are on the same scale, which is required for LASSO selection. The LASSO regression is then run on the standardized residuals

of the dependent variable and the independent variables. The LASSO regression selects a subset of weather variables that are most relevant for predicting economic growth. Finally, we estimate Equation 1 using the selected weather variables.

We deviate from Akyapi et al. (2024) in that we perform the selection with Adaptive LASSO instead of using the Bayesian Information Criteria LASSO. Adaptive LASSO is preferable to cross-validation (CV) and Bayesian Information Criterion (BIC) LASSO. By assigning weights to penalize coefficients based on initial consistent estimators, addresses the their tendency to over-penalize significant predictors while insufficiently penalizing irrelevant ones. This methodological refinement yields the oracle property, whereby the procedure asymptotically performs as if the true model were known a priori, something that is not guaranteed by CV LASSO or BIC LASSO. Empirical evidence from applications to panel data and instrumental variable specifications demonstrates that adaptive LASSO produces more parsimonious models with improved out-of-sample performance, particularly in high-dimensional settings with complex economic relationships. The Post-LASSO regressions are weighted by population to provide more representative estimates of the average treatment effect of weather extremes on economic growth.

4 Results

The inclusion of several climate statistics provides a more nuanced understanding of the relationship between weather extremes and economic growth. The results for Europe are presented in Table 1. Coefficients represent the conditional effect on growth of a one standard deviation increase in the weather variable. Unstandardized coefficients are reported in Appendix A. Shrinkage selects overlapping but different sets of regressors for GRP and its sectoral equivalents. Note that we do not report standard errors for the Post-LASSO estimates since they are invalid for inference. They are inconsistent because they condition on a fixed set of covariates when those covariates were actually chosen using the data. This selection process invalidates the classical distributional theory. However, Post-LASSO coefficients are valid for prediction tasks.

Gross Regional Product For aggregate gross regional product (GRP), the Adaptive LASSO procedure selects three climate variables: temperature variance (TVAR), wet days (WD), and average daily minimum temperature (TN). All three coefficients are negative. Average minimum temperature shows the largest effect (-0.020), indicating that years with unusually warm nighttime temperatures predict lower economic growth. Temperature variance also negatively predicts growth (-0.011), consistent with Linsenmeier (2023), who finds that temperature volatility reduces economic activity beyond what can be attributed to mean temperature shifts. A higher number of wet days is associated with modestly lower growth (-0.006).

Agriculture The agricultural sector responds to a distinct set of climate variables: heatwaves (HW), total precipitation (RR), and drought (SPEI). Notably, this is the only sector for which the LASSO procedure does not select temperature-level variables (TN, TVAR), instead retaining only extreme-event indicators. Total precipitation has the largest negative coefficient (-0.051), suggesting that excess rainfall harms agricultural output, possibly through waterlogging or flooding. Drought as measured by SPEI also reduces agricultural growth (-0.023), indicating that both ends of the moisture spectrum are detrimental. Heatwaves negatively affect agricultural output (-0.022), consistent with crop stress during extreme heat. Agriculture has the highest adjusted within- R^2 (3.4%), confirming that weather variation explains a larger share of output fluctuations in this sector than in any other.

Manufacturing Manufacturing displays the most parsimonious specification, with only one selected variable: average daily minimum temperature (TN), with a coefficient of -0.032 . The absence of precipitation-related, heat-related, or temperature-volatility variables suggests that, conditional on fixed effects and trends, the main weather channel affecting European manufacturing is nighttime warmth, which may reduce labor productivity through impaired sleep recovery (Zivin et al., 2025). The adjusted within- R^2 is the lowest across sectors (0.23%), indicating that weather variation has limited predictive power for manufacturing growth in Europe.

Services The services sector shows the broadest set of selected variables (five): average daily minimum temperature (TN), temperature variance (TVAR), heatwaves (HW), extreme precipitation (RX), and cold waves (CW). Minimum temperature (-0.041) and temperature variance (-0.019) have the largest negative effects, mirroring the pattern observed for GRP and consistent with Linsenmeier (2023). Given the dominant weight of services in European economies, these effects are likely the main drivers of the aggregate GRP results in Column (1). Cold waves also negatively predict services growth (-0.004). Interestingly, heatwaves ($+0.007$) and extreme precipitation events ($+0.004$) carry small positive coefficients, which may reflect increased demand for cooling, hospitality, or repair services following extreme events. The adjusted within- R^2 (1.4%) is the second highest after agriculture, indicating meaningful weather sensitivity for the tertiary sector.

Table 1: Post-LASSO regression results for the growth rate of gross regional product per capita and its sectoral equivalents for Europe. Coefficients are standardized.

	(1) GRP	(2) Agriculture	(3) Manufacturing	(4) Services
Standardized TVAR	-0.01109			-0.01938
Standardized WD	-0.00561			
Standardized TN	-0.01978		-0.03230	-0.04098
Standardized HW		-0.02190		0.00674
Standardized RR		-0.05119		
Standardized SPEI		-0.02250		
Standardized RX				0.00376
Standardized CW				-0.00384
N	1.02e+04	8.92e+03	8.92e+03	8.90e+03
Adjusted R2	0.64656	0.32806	0.55360	0.62906
Adjusted within-R2	0.00345	0.03407	0.00233	0.01432
BIC	-2.83e+04	-9.86e+03	-1.89e+04	-2.33e+04

Notes: The table presents the results of the Post-LASSO regression. The dependent variable is the growth rate of gross regional product or of its sectoral equivalent. The independent variables are weather extremes and have been selected through Adaptive LASSO. All regressions include region fixed effects, year fixed effects, and regional linear time trends.

5 Conclusions

Our empirical analysis reveals that the relationship between weather extremes and economic growth varies substantially across sectors in European regions. The heterogeneity in sectoral responses challenges the conventional focus on mean temperature as the sole climate proxy, highlighting the importance of considering a broader range of climate variables when modeling economic impacts. Importantly, the convex temperature-growth relationship observed in European regions contrasts

with the global concave pattern documented in prior literature, suggesting that regional climate adaptation strategies require tailored approaches.

These findings have significant implications for climate policy and economic modeling. First, they demonstrate the value of sector-specific climate damage functions that capture the complex, non-linear responses to various weather extremes. Second, they underscore the importance of sub-national analysis in revealing heterogeneous impacts that may be obscured in country-level studies. Third, as climate change intensifies extreme weather events, sector-specific damage functions provide a more nuanced foundation for adaptation planning and cost-benefit analyses of climate policies.

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A Appendix A

A.1 Weather extremes

Table A1: Weather extremes used in the analysis.

Indicator	Definition
<i>HW</i> (Heatwave)	At least five consecutive days with $TX > 90\text{th percentile}$ (vs. 1981-2010) and $TX \geq 30^\circ\text{C}$. Computed over the warmest six months: Apr-Sep for the Northern Hemisphere and Oct-Mar for the Southern Hemisphere (including the next year).
<i>CW</i> (Coldwave)	At least five consecutive days with $TN < 10\text{th percentile}$ (vs. 1981-2010) and $TN < 0^\circ\text{C}$. Computed over the coldest six months: Oct-Mar for the Northern Hemisphere and Apr-Sep for the Southern Hemisphere (including the next year).
<i>PET</i> (Potential Evapotranspiration)	Computed using Hargreaves-Samani's equations.
<i>PEXT</i> (Precipitation?)	Total precipitation over the year.
<i>RR</i> (Total Precipitation)	Total precipitation over the year.
<i>RX</i> (Extreme Precipitation)	Maximum 5-day precipitation cumulate over the year.
<i>SPI, SPEI</i> (Drought)	A drought event starts when the indicator (12-month accumulation) falls below -1 for at least 2 consecutive months and ends when it turns positive again.
<i>SNX</i> (Snowfall)	Snowfall amount over the year.
<i>TM</i> (Mean Temperature)	Yearly average of daily mean temperature.
<i>TN</i> (Minimum Temperature)	Yearly average of daily minimum temperature.
<i>TX</i> (Maximum Temperature)	Yearly average of daily maximum temperature.
<i>TVAR</i> (Temperature Variance)	Yearly variance of daily mean temperature.
<i>WD</i> (Wet Days)	Number of days with precipitation $> 1\text{ mm}$.

A.2 Post-LASSO regression results - Europe (unstandardized coefficients)

Table A2: Post-LASSO regression results for the growth rate of gross regional product per capita and its sectoral equivalents for Europe. Coefficients are not standardized.

	(1) GRP	(2) Agriculture	(3) Manufacturing	(4) Services
TVAR	-0.0154			-0.0270
WD	-0.000154			
TN	-0.00698		-0.0114	-0.0145
HW		-0.00145		0.000446
RR		-0.000160		
SPEI		-0.00466		
RX				0.000133
CW				-0.000137
N	10235	8916	8919	8903
Adjusted R2	0.647	0.328	0.554	0.629
Adjusted within-R2	0.00345	0.0341	0.00233	0.0143
BIC	-28338.8	-9857.4	-18854.1	-23265.2

Notes: The table presents the results of the Post-LASSO regression. The dependent variable is the growth rate of gross regional product or of its sectoral equivalent. The independent variables are weather extremes and have been selected through Adaptive LASSO. All regressions include region fixed effects, year fixed effects, and regional linear time trends.

A.3 Post-LASSO regression results - Global (standardized coefficients)

Table A3: Post-LASSO regression results for the growth rate of gross regional product per capita and its sectoral equivalents for the world. Coefficients are standardized.

	(1) GRP	(2) Agriculture	(3) Manufacturing	(4) Services
Standardized HW	0.00222	-0.01585		
Standardized PET	-0.11329			
Standardized RR	0.01443	-0.02816		
Standardized SNX	-0.00626			
Standardized TVAR	-0.05369	-0.01219	-0.03125	-0.03407
Standardized WD	-0.02722			-0.02173
Standardized SPEI	-0.00937	-0.01132	-0.00291	-0.00439
Standardized CW	-0.00331	-0.00316	-0.00390	-0.00497
Standardized TN	-0.18569	0.06248		
Standardized TX	0.22818			0.03624
Standardized RX		-0.00443		
N	4.37e+04	3.32e+04	3.33e+04	3.33e+04
Adjusted R2	0.27313	0.18203	0.27298	0.35270
Adjusted within-R2	0.01183	0.01513	0.00236	0.00787
BIC	-6.81e+04	-3.15e+04	-4.03e+04	-6.38e+04

Notes: The table presents the results of the Post-LASSO regression. The dependent variable is the growth rate of gross regional product or of its sectoral equivalent. The independent variables are weather extremes and have been selected through Adaptive LASSO. All regressions include region fixed effects, year fixed effects, and regional linear time trends.

A.4 Post-LASSO regression results - Global (unstandardized coefficients)

Table A4: Post-LASSO regression results for the growth rate of gross regional product per capita and its sectoral equivalents for the world. Coefficients are not standardized.

	(1) GRP	(2) Agriculture	(3) Manufacturing	(4) Services
HW	0.000158	-0.00113		
PET	-0.000318			
RR	0.0000156	-0.0000305		
SNX	-0.000467			
TVAR	-0.0451	-0.0102	-0.0263	-0.0286
WD	-0.000420			-0.000336
SPEI	-0.00209	-0.00253	-0.000651	-0.000981
CW	-0.000136	-0.000130	-0.000161	-0.000204
TN	-0.0260	0.00875		
TX	0.0304			0.00482
RX		-0.0000752		
N	43653	33239	33325	33310
Adjusted R2	0.273	0.182	0.273	0.353
Adjusted within-R2	0.0118	0.0151	0.00236	0.00787
BIC	-68104.9	-31481.6	-40284.5	-63790.6

Notes: The table presents the results of the Post-LASSO regression. The dependent variable is the growth rate of gross regional product or of its sectoral equivalent. The independent variables are weather extremes and have been selected through Adaptive LASSO. All regressions include region fixed effects, year fixed effects, and regional linear time trends.

A.5 Quadratic damage functions

Table A5: Regression results for the growth rate of gross regional product per capita and its sectoral equivalents for Europe, including a limited set of weather statistics. Coefficients are standardized.

	(1) GRP	(2) Agriculture	(3) Manufacturing	(4) Services
TM	-0.0235 (0.015)	-0.0324** (0.014)	-0.0334 (0.020)	-0.0064 (0.019)
TM $\times$ TM	0.0013** (0.001)	0.0002 (0.001)	0.0013 (0.001)	0.0002 (0.001)
RR	0.0000 (0.000)	0.0001* (0.000)	0.0000 (0.000)	-0.0001 (0.000)
RR $\times$ RR	-0.0000 (0.000)	-0.0000*** (0.000)	-0.0000* (0.000)	0.0000 (0.000)
N	9958.0000	8811.0000	8814.0000	8806.0000
Adjusted R2	0.6282	0.3450	0.5947	0.0656
Adjusted within-R2	0.0023	0.0065	0.0034	0.0000

Notes: The dependent variable is the growth rate of gross regional product or of its sectoral equivalent. All regressions include region fixed effects, year fixed effects, and regional linear time trends.